

REPUBLIC OF LEBANON

COUNCIL FOR DEVELOPMENT AND RECONSTRUCTION

DETAILED ENGINEERING DESIGN

FOR THE CONSTRUCTION OF

AKKAR GOVERNMENTAL HOSPITAL

TERMS OF REFERENCE

APPENDIX A - Construction

HARBATA GOVERNMENTAL HOSPITAL

TERMS OF REFERENCE

1. DESCRIPTION OF THE PROJECT

The aim of the project is to carry out the studies for the construction works of Akkar Governmental Hospital as a One Day Surgery Hospital.

1.1. Function

The major and required functions covers the following services :

- Image department
- Outpatient clinics
- Laboratory department
- One-day surgery operating room
- Emergency department

1.2. Surface Program

According to the medical program the Consultant shall establish an area program in coordination with the MOH.

1.3. Budget

The allocated budget is equivalent to /3,000,000/ US\$ for the construction works. The budget allocated for the medical equipment and furniture is approximately equivalent to an amount of /1,000,000/ US\$.

1.4. Land

The plot of land is located in Sahleh (Akkar District) and it belongs to the Municipality of Sahleh.

2. SCOPE OF WORK :

The services required will be to design the Hospital building to accommodate the medical program in consideration with the Client and according to the Ministry of Public Health requirement.

3. DEFINITION OF THE CONSULTANCY

The following duties shall be carried out :

3.1. Design

3.1.1. Consultancy duties

Perform all architectural and engineering studies as described herein, with due care and diligence to obtain the objectives of this appointment.

3.1.2. Site Survey

Within the scope of this appointment, cause to be conducted all topographical and geotechnical surveys, that will be required for the preparation of the preliminary and detailed engineering design of the project components.

3.1.3. Design

Prepare preliminary and detailed designs, specifications, bills of quantities and tender documents for all components of projects.

3.1.4. Assistance of the Client

The Client will provide the consultant with available information, data and reports relating to these services. The Consultant shall be responsible, however, for the analysis and interpretation of the data received for his conclusions and recommendations.

3.1.5. Responsibility of data correctness

All information, data and reports obtained from the Client shall be reviewed, completed as necessary, and analyzed by the consultant. The responsibility for the correctness of the said data shall rest with the consultant.

3.1.6. Standards

The design of the project components shall be prepared in accordance with the international standards for medical facilities as regards to function, stability, security, aesthetics and environmental considerations.

3.1.7. Lebanese regulations

The design shall conform to the regulations, laws, by-laws and codes of practice in force in the Lebanese Republic with particular emphasis on resistance to earthquakes, fire protection and on the safety of persons using the facilities.

3.1.8. Consultant performance

The Consultant shall delegate skilled persons for all his professional acts and mobilize the appropriated means to perform these obligations within the duration agreed here after.

3.2. Site Supervision : N.A.

4. PHASES OF THE MISSIONS

Duties described hereunder shall be carried out according to the phases of the studies with succession and overlapping periods which shall be defined in the general time schedule. The organization of the phases shall be as follows

4.1. Phase 1 :

Investigation, surveys, finalization of the medical program with MOH.

4.2. Phase 2 :

Preparation of the conceptual design and preliminary listing of the major medical equipment.

4.3. Phase 3 :

Preparation of the permit documents.

4.4. Phase 4 :

Preparation of the preliminary detailed design and loaded drawings

4.5. Phase 5 :

Preparation of the detailed final design, specifications, bills of quantities and tender documents.

4.6. Phase 6 :

Assistance to the Client during the tender phase.

5. PHASE 1: INVESTIGATION SURVEYS AND FINALIZATION OF THE MEDICAL PROGRAM

The Consultant shall proceed to the following investigations:

5.1. Visual Survey

Visit of the location of the project and formulate an idea of the site, its topography, access, and proximity to surrounding functions.

5.2. Data collection:

Collect data pertaining building codes and constraints, existing and planned roads, and infrastructure services related to water, sewer, power, telephone, etc..

5.3. Topographic survey:

Prepare the topographic survey tender documents (contract to be financed by the Client). The Consultant task includes the supervision of the relative works.

5.4. Geotechnical investigations :

Prepare of the soil investigation tender documents for geotechnical investigations and tests that are necessary to determine the necessary data and criteria for the building design. Consultant task includes the interpretation of the concerned report; (Contract to be financed by the client).

5.5. Medical Program:

Prepare the necessary documents in order to illustrate the result of the feasibility study in what is in relation with the medical program and needs. Develop this program in close coordination with the Client and MOH, define the number of floors and the built-up areas based on prevailing Building codes.

5.6. Deliverables

Report :

At this end of this phase the deliverables are :

- Report including the description of the site visit, the collected codes and construction regulations, إفادة عقارية، إرتفاق وتخطيط و خريطة مساحة.

- Tender documents related to the topographical and geotechnical site surveys.
- The detailed medical program and needs.
- The illustration of the built-up area and the distribution of the function on the different floors.
- A preliminary cost estimate.

6. PHASE 2 : CONCEPTUAL DESIGN AND PRELIMINARY LISTING OF THE MAJOR MEDICAL EQUIPMENT.

On the basis of the studies approved by the client and as the case may be, taking into consideration the modification required by the Client, the Consultant shall perform the conceptual design.

6.1. Conceptual design

Prepare a conceptual design of the medical facility and produce in sufficient details for presentation with the building program to the Client for discussion and approval.

Taken into consideration the future extension of the Hospital without any cut of the existing services.

6.2. Preliminary listing of the medical equipment.

Establish a preliminary list of the proposed major medical equipment in order to ensure that the allocated spaces can house the requirement of the medical program.

6.3. Standard

The design of the project shall be prepared in accordance with international standards for functions and health and safety environment.

The construction shall be verified to the Para seismic calculation, and the regulation for the fire protection and the safety of the persons shall be applied.

6.4. Optimization

The consultant shall program functional relationships that shall reduce time and effort, eliminate cross traffic and reduce the need for costly hard to maintain automated devices

6.5. Program definition:

Carry out the necessary studies and comparative analysis to determine the various solutions for accommodating the medical programs on the project sites within the allocated project budget.

Develop detailed programs in close cooperation with the Client, identify major functions and their related services, define number of floors built-up areas based on prevailing building codes.

6.6. Environmental impact

Investigate and assess the possible environmental impact of each of the project's component.

6.7. Technical report

Prepare a technical report where the consultant should explain the idea and the concept, for the design criteria adopted for the structure mechanical, electrical, medical equipment, architecture etc....

The standard used in the design, the type of the finishes and their location and any other details.

6.8. Cost estimate

Based on the above information prepare a preliminary cost estimate. The cost estimate will be budgetary.

6.9. Deliverables

Graphic documents :

- General site plan including building layout and orientation, access road external existing services proposed services connections, external facilities, fencing and landscaping, scale 1/200
- Floor plan drawings for each level : (showing the provision for the future extension) scale 1/100
- Floor plan drawings of each section : scale 1/50
- Rendered elevations scale 1/100
- General sections scale 1/100
- Perspective artist impression
- Vertical and Horizontal circulation flow diagrams : Patient access, visitor access, services access, car access, ...

Report

- Schedule of areas per floor, per section and total built-up areas.
- Comparison table between the built and allowable constructed area.
- Environmental assessment.
- Technical explanation of all trades.
- Cost estimation (budgetary).

7. PHASE 3 : PERMIT DOCUMENTS :

After the approval of the Client on the preliminary design project, the Consultant shall develop :

Documents :

The necessary document for the building permit as required by the Local regulation: "Ordre des Ingénieurs" and Town planning department ", Civil defense and Electricity authorities.

Follow up :

The Consultant will be responsible for the following up the procedures for the permit or other licenses necessary for the implementation of the project. He shall provide all the information details or documents required by the authorities in order to obtain the permit at the soonest.

Authorities requirements

The consultant is responsible for preparing all complementary documents or works requested by the authorities for obtaining the permits necessary for the execution of the project (alignments, etc ...)

Consultant responsibility

The consultant will be responsible for the correctness of any information or document presented to the authorities, and shall deliver the same to the relevant authorities under his responsibility.

Obtention of permit

The consultant will be responsible for obtaining and submitting to the Client all the necessary permits, approvals and licenses needed to complete the implementation of the medical facilities

Progress report :

The Consultant shall report to the client on the progress of the permits procedures, and shall consult with the client regarding any envisaged delay in obtaining the necessary permits. The Client shall make all possible effort to assist in expediting the procedures

7.1. Deliverables

Report :

Evidences document confirming the submittal of the documents to the "Order des Ingénieurs".

8. PHASE 4: PRELIMINARY DETAILED DESIGN AND LOADED DRAWINGS

Based on the conceptual design, the client's comment thereon, and its modifications, develop the design to preliminary stage including such addition details as may be necessary to define and size all the project component. The documents shall include :

Architectural drawings :

- Floor plans indicating the room numbers, the finishing schedules and colors schemes scale 1/100
- Elevations scale 1/100
- Sections : scale 1/100
- Reflected ceiling plans scale 1/100
- Typical wall sections scale 1/20
- Typical details : Roofing, waterproofing, ... scale 1/20
- Fire zoning escape and fire routing scale 1/100
- Fire strategy
- Traffic design for the elevators.

Structural drawings :

- Framing structural drawings and its approximate dimensions.
- Foundations drawings.

Electrical drawings :

- Schematic design and risers drawings : low and extra low voltage drawings
- The traffic design for the elevator

Mechanical drawings :

- Schematic design and riser drawings : Plumbing heating, air conditioning, ventilation, medical gazes, fire protection.

Loaded drawings

- Medical and non medical equipment and furniture loaded drawings for all floor showing the electromechanical requirements scale 1/100

8.1. Deliverables

Graphic documents :

- Drawings of the above listed trades.

Report :

- Design criteria, basic data and codes
- Description of the proposed different mechanical, electrical and medical system and functions
- Breakdown of the constructed area with the cost estimate

9. PHASE 5 : DETAILED FINAL DESIGN, SPECIFICATIONS AND TENDER DOCUMENTS :

Based on the preliminary design, the Client's comment and its modifications, the Consultant shall develop the detailed final design, specifications and tender documents including :

Architectural drawings

- | | |
|--|-----------------|
| • Master Plan : | 1/200 |
| • Floor plan drawings of each level : | 1/100 |
| • Elevations : | 1/100 |
| • Sections : | 1/100 |
| • Wall sections : | 1/20 |
| • Reflected ceiling : | 1/100 |
| • Stairs details : | 1/20 |
| • Lifts details : | 1/20 |
| • Toilet details | 1/20 |
| • Roofing waterproofing | 1/20 |
| • External work details | 1/50-1/20 |
| • Main entrance and fence details | 1/100-1/50-1/20 |
| • Landscaping plan, list of plants : | 1/100 |
| • Medical loaded drawings | 1/100 |
| • Aluminum windows schedule | --- |
| • Timber door schedule | --- |
| • Steel door schedule | --- |
| • Signage and furniture drawings of each level | 1/100 |
| • X-Ray rooms and protection | |

Structural drawings :

- | | |
|--------------------|-------|
| • Setting out plan | 1/200 |
| • Excavation plan | 1/100 |
| • Foundation plan | 1/100 |

- Ground slab floor formworks and reinforcement 1/100
- Formwork and reinforcement plan of each level 1/100
- Stairs details 1/20
- Foundation schedule and details 1/50-1/20
- Tie beam schedule and details 1/50-1/20
- Column schedule and details 1/50-1/20
- Beams details 1/50-1/20
- Rib details 1/50-1/20
- Lift core details 1/50-1/20
- Water tank details 1/50-1/20
- Calculations notes (Robot)

Electrical drawings : Low and extra low voltage

- Floor plan drawings 1/100
- (Power, lighting , nurse call, IT, data system, public address, fire alarm, external lighting, protection system, CCTV, TV, internet, BMS)
- Riser diagram ---
- Panel board schedule ---
- Details
- Left and conveyor

Mechanical drawings : Drainage water supply, fire fighting, irrigation and medical gazes

- Floor plan drawing of each level : 1/100
- Riser diagram ---
- Technical room ---
- Water tank ---
- Treatment plant ---
- Acoustic design (drawings and calculation notes)

Heating, ventilation and air conditioning : piping and ducting

- Floor plan drawing of each level 1/100
- Riser diagram ---
- Technical room ---
- List of Equipment ---

Technical specifications, bills of Quantities

They include :

- The detailed specifications defining without ambiguity in accordance with the detailed drawings, the works of the various specialties.
- The technical specifications and the workman ship of the different materials and systems.
- Schedule of doors and ironmongery
- Last of finishes by room
- Itemized bill of quantities with the unit price list enumerating the various type of material used.
- Priced BOQ.
- Tender documents including :
 - Full set of drawings with graphic and written detail
 - General and special specifications, bills of Unit prices and bill of Quantities detailing all the component of the works and all items of supplies and installation.
 - General conditions of contract, Particular Conditions of Contract, form of the Contract and form of tender.

9.1. Deliverables

Graphic documents :

- The detailed execution drawings.

Documents :

- Specifications, bills of Quantities, tender documents.

10. Phase 6 : TECHNICAL ASSISTANCE DURING TENDER PROCEDURE

The Consultant shall assist the Client in the Tender procedure to prepare: clarification for Tenderers, evaluation report and the contract documentation for signature by the parties.

10.1 Tender Period

The consultant shall submit to the Client the tender documents for issuing. He shall not distribute any additional copies (even partial) or information without the written comment of the Client.

10.2 Answer to bidders :

During tender period, the Consultant shall provide the tenderers with precision and complementary information they required, only according to the Client specific instructions:

10.2.1 Bidders meeting:

The Consultant shall participate with Client's representatives to the bidder meeting when foreseen in the bidding documents. He will draft and submit to the Client the minutes of the meetings.

10.2.2 Bidder's queries:

The Consultant shall propose to the Client answers to the query raised by any of the contractors invited to tender for the project as part of it. No query can be answered directly by the Consultant but only by the Client and to all the bidders.

10.3 Bid opening and bid analysis

At the submission deadline, the Client designates an evaluation committee assisted by a consultant representative to proceed with the bids opening and bids analysis. Preparation of computed data sheet for record.

The evaluation committee first check the conformity of the documents attached to the bids.

Then the offers shall be analyzed.

The consultant shall establish, on the basis of the computed forms for bills of unit price and bills of quantities the following documents:

- 10.3.1** Comparison table of the unit prices between all bidders and the cost estimate, highlighting the better prices and the missing or wrong information.
- 10.3.2** From the unit prices applied to the bill of quantities, the Consultant shall verify the computed offer submit by the bidders and indicate any bidder's mistake.
- 10.3.3** The Consultant shall report to the Client the result of these prices comparison and issue draft letter for each bidder specifying and explaining the discrepancies in their offer.
- 10.3.4** All these tasks shall be performed in Client offices in a total confidentiality. Any and all information shall be kept between the only members of the evaluation committee.

10.4 Contract preparation

As soon as the evaluation report is approved by the Client and donors the Consultant shall assist the Client in preparing the contracts for each retained company, the contract shall include all the contract documents as per the bidder offer and including the results of negotiations if any.

11. STAFF

The Consultant shall employ the staff as may be necessary to fulfill his obligation under the agreement as specified in Appendix B.

12. NUMBER OF COPIES :

Reports :

- Unless otherwise stated in the document. Report shall be submitted to the Client in 2 copies.

Graphic documents :

- Unless otherwise stated in this documents, graphic documents shall be submitted to the Client as follows :
 - For phases 1, 2, 4 : One hard copy + one copy for the MOH approval where requested.
 - For phase 3 : As per the regulation
 - For phase 5 : Two hard copies.
 - Final package for bid purpose : Five hard copies + 1 soft copy

13. TIME FRAME :

The time frame needed to implement the above work is 6 months, subdivided as follows:

- Phase 1 : ND + 1
- Phase 2 : ND + 2
- Phase 3 : Phase 2 approval +1 (based on the concerned administrative)
- Phase 4 : Phase 2 approval + 2
- Phase 5 : Phase 4 approval + 1

1. NUMBER OF COPIES:

The documents related to phases I, II and III will be submitted in two hard copies and one soft copy.

APPENDIX B

Consultant 's Personnel

The consultant shall employ such staff as may be necessary to fulfill his obligations under the agreement. The consultant shall make his own assessment of the staff needed for carrying out the work but must comprise at least the minimum specified here:

Project Manager/ Architect
Structure Engineer
Senior Biomedical engineer
Electrical and Mechanical engineers/ or Electro-Mechanical Engineer
Additional staff or specialists (Biomedical engineers, technicians,...) as needed.

a. Project manger:

Principal duties: represents the consultant to the client. Holds overall responsibility for the direction of the consultant's team. Maintains contact with the client and the consultant main office. Liaises with all parties concerned in the implementation of the project. Maintains control of programming and implementation and ensures that works are proceeding according to schedule.

Qualifications: University degree in Architecture with a minimum of 12 years of experience since graduation with at least design of one hospital project. He should have been a permanent staff in the office of the consultant since at least one year.

b. Civil Engineer (Structure Engineer)

Principal duties: responsible for the structure studies of the building. ensures that all international regulations are respected. Make sure that the building is designed to seismic norms according to the zone where the hospital is situated.

Qualifications: University degree in civil engineering with a minimum of 10 years of experience since graduation with at least structural design of one hospital project or similar complexity project.

c. Senior Biomedical engineer (or Biomedical firm):

Principal duties: ensures that all the equipment are conform with the hospital needs, check the architecture drawing to insure that areas can receive the equipment properly and to prepare the documents mentioned in the TOR.

Qualifications: University degree in the specialty with at least 10 years of experience. Should have worked in at least one project of similar nature (design of installation of Medical Equipment & Furniture in a newly built or rehabilitated or extended hospital) in the past 10 years.

d. Electrical engineer

Principal duties: In addition to design task he should Coordinates with the biomedical engineer to ensure that all needed connections required for the adequate installation of the equipment are in place. This shall be planned before the end of the phase 4 period and shall ensure the needed step for the execution of works.

Qualifications: University degree in the field with at least 10 years of experience in building projects and at least three years of experience in the hospital sector. Should have worked in at least one hospital project of similar nature within the last 10 years.

e. Mechanical Engineer

Principal duties: In addition to design task he should Coordinates with the biomedical engineer to ensure that all needed connections required for the adequate installation of the equipment are in place. This shall be planned before the end of the phase 4 period and shall ensure the needed step for the execution of works.

Qualifications: University degree in the field with at least 10 years of experience in building projects and at least three years of experience in the hospital sector. Should have worked in at least one project of similar nature within the last 10 years.

The consultant shall express his staff list supporting the project team in management of the mission.

f. Client authorization

Any staff allocated to the project should have the client (CDR) authorization, the client have the right to refuse any staff, and the consultant should replace it within 15 days.

APPENDIX C

Financial details : Consultant's fees

A – Construction

The consultant shall present a work plan for each task of the services.

The remuneration will be considered as a lump sum, detailed according to a schedule that should show as a price analysis so that the rate of every task is seen clearly.

A -1 Design Fees

Design Fees	Lump sum		US
Tender Evaluation	Lump sum		US
Total without VAT			US

In letters _____

A – 2 Method of payment

Phase 1	ND + 1	10%	Upon approval
Phase 2	Phase 1 + 1	30%	Upon approval
Phase 3	Phase 2 + 1	10%	Upon Submission to Order of Engineers
Phase 4	Phase 2 + 3	30%	Upon approval
Phase 5	Phase 4 + 1	20%	Upon approval
		100%	
Phase 6 Tender Evaluation		L.S.	Upon approval of committee report

A – 3 PERMIT COST

Any taxes, fees, stamp duties “Ordre des Ingénieurs” charges or taxes, or any other municipality or governmental authorities’ taxes shall be paid directly by the Client and are not considered within the fees amount of the Consultant.

A – 4 COST OF SITE SURVEY AND GEOTECHNICAL INVESTIGATION REPORT

All charges to subcontractors to provide site survey and geotechnical investigation report as stated in phase 1 of the design shall be paid directly by the client and are not considered within the fees amount of the consultant.